

Vendor Performance Analysis Project Report

Business Problem:

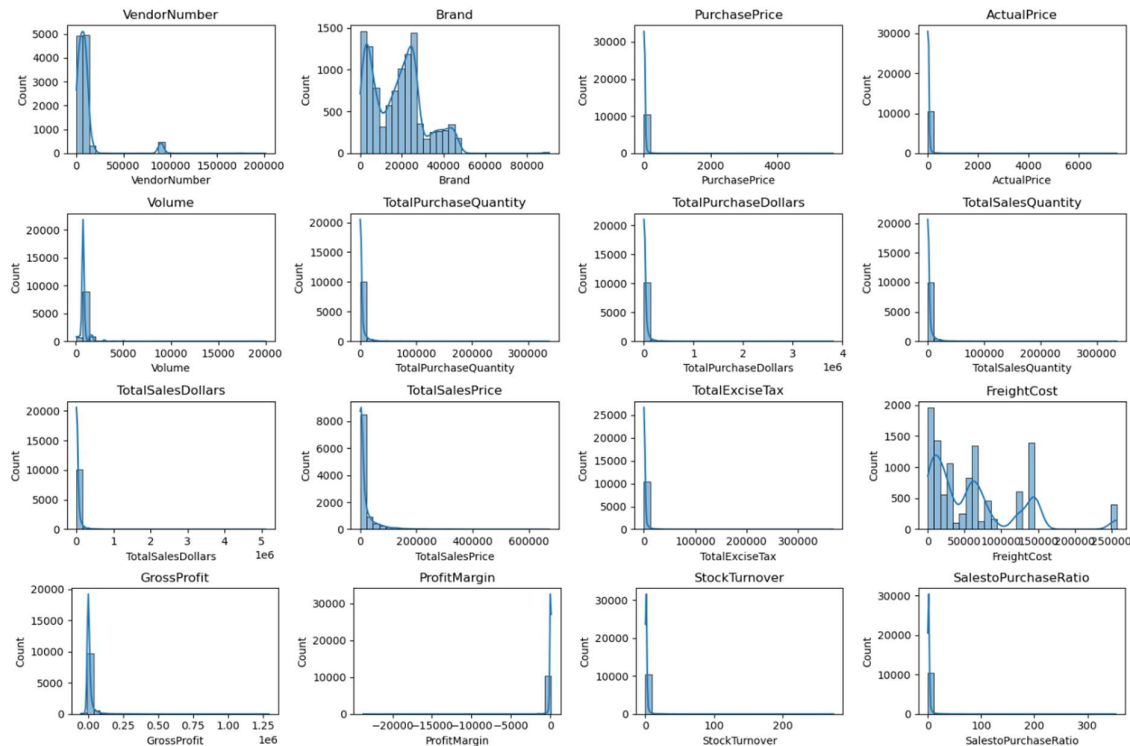
Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

Exploratory Data Analysis Insights:

Summary Statistics –

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalestoPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02



Negative & Zero Values:

- **Gross Profit:** Minimum value is -52,002.78, indicating losses. Some products or transactions maybe selling at a loss due to high costs or selling at discounts lower than the purchase price.
- **Profit Margin:** Has a minimum of -inf, which suggests cases where revenue is zero or lower than costs.
- **Total Sales Quantity & Sales Dollars:** Minimum values are 0, meaning some products were purchased but never sold. These could be slow-moving or obsolete stock.

Outliers Indicated by High Standard Deviation:

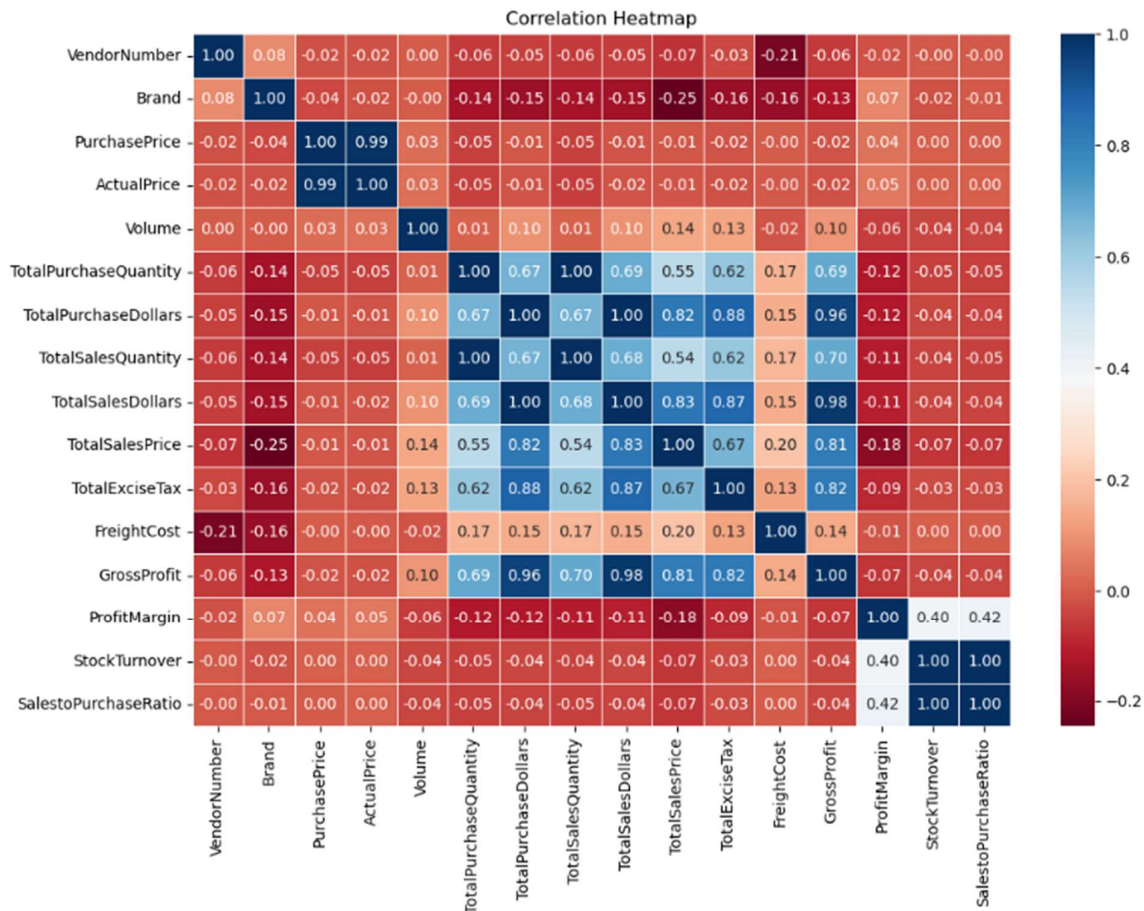
- **Purchase & Actual Prices:** The max values (5,681.81 & 7,4999.99) are significantly are higher than the mean (24.39 & 35.64), indicating potential premium products.
- **Freight Cost:** Huge variation from 0.09 to 257,032 suggests logistics inefficiencies or bulk shipments.
- **Stock Turnover:** Ranges from 0 to 274.5 indicating that some products sell extremely fast while others remain in stock indefinitely.

Data Filtering –

To enhance reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses)
- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions)
- Sales Quantity = 0 (to eliminate inventory that was never sold)

Correlation Insights –



Purchase Price vs. Total Sales Dollars: Purchase Price has weak correlation with Total Sales Dollars, suggesting price variations do not significantly impact sales or revenue.

Purchase Price vs. Gross Profit: Purchase Price has weak correlation with Gross Profit, suggesting that Gross Profit is not affected by pricing variations.

Total Purchase Quantity vs. Total Sales Quantity: Total Purchase Quantity and Total Sales Quantity are perfectly correlated (1.00). This suggests very little

unsold inventory in the aggregated data; nearly every purchased item is eventually sold.

Profit Margin vs. Total Sales Price: Profit Margin exhibits a weak negative correlation (-0.18) with Total Sales Price, indicating that higher-priced sales are associated with slightly lower margin percentages. This may reflect competitive pricing, bulk discounts, or lower percentage markups on premium products.

Stock Turnover vs. Gross Profit & Profit Margin: Stock Turnover has a negative correlation with Gross Profit (-0.04) and a moderate correlation with Profit Margin (0.40), indicating that faster turnover does not necessarily result in higher profits.

Research Questions & Key Findings:

1. Brands for promotional or Pricing Adjustments –

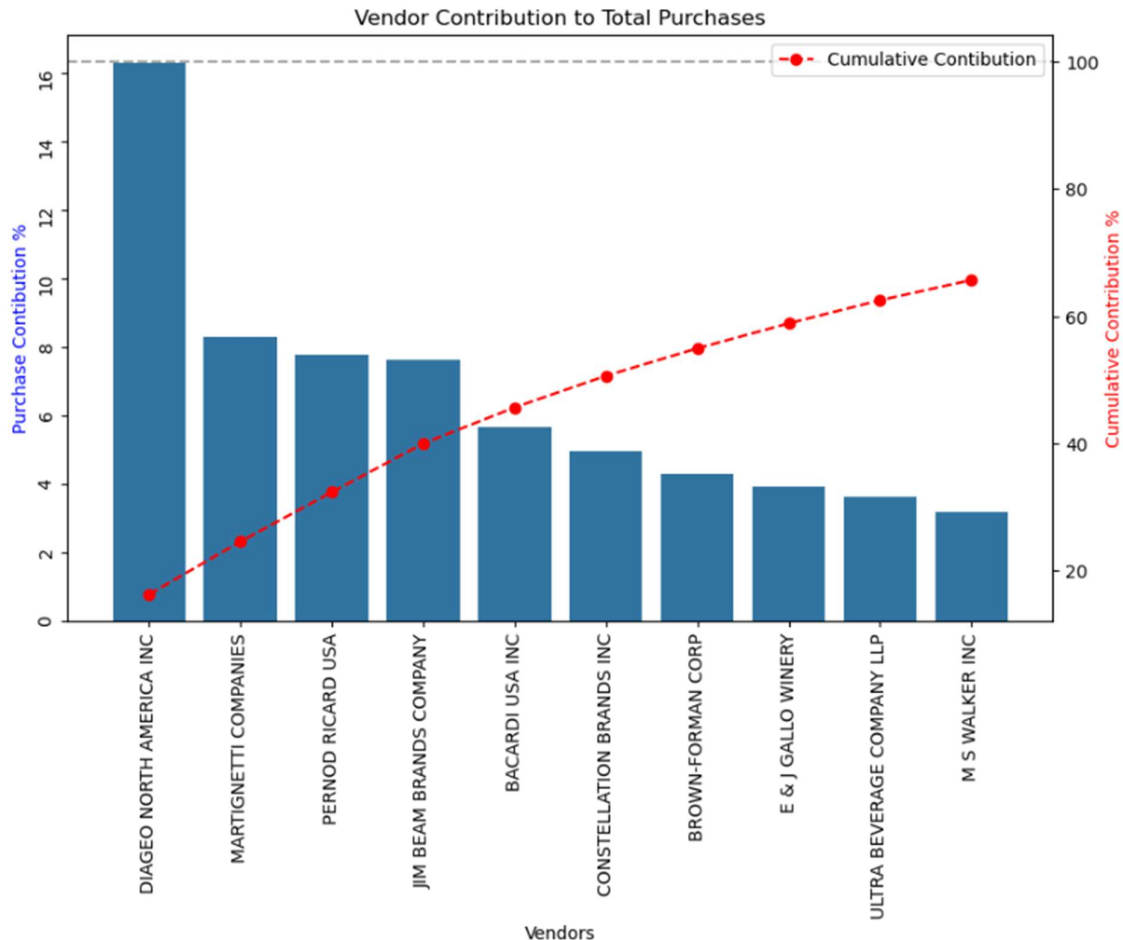
Brands who need Promotional or Pricing Adjustments:			
	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournal 10	558.87	69.434752

198 rows × 3 columns

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.

2. Top Vendors by Sales & Purchases Contribution –

The top 10 vendors contribute approximately 65% of total purchases. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings –

Vendors buying in large quantities receive a 72% lower unit cost (\$10.78 per unit vs. higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

<u>Order Size</u>	<u>Unit Purchase Price</u>
Small	39.068186
Medium	15.486414
Large	10.777625

4. Identifying Vendors with Low Inventory Turnover –

Total Unsold Inventory Capital: \$2.71M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability. Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

StockTurnover	
VendorName	
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

5. Profit Margin Comparison: High vs. Low Performing vendors –

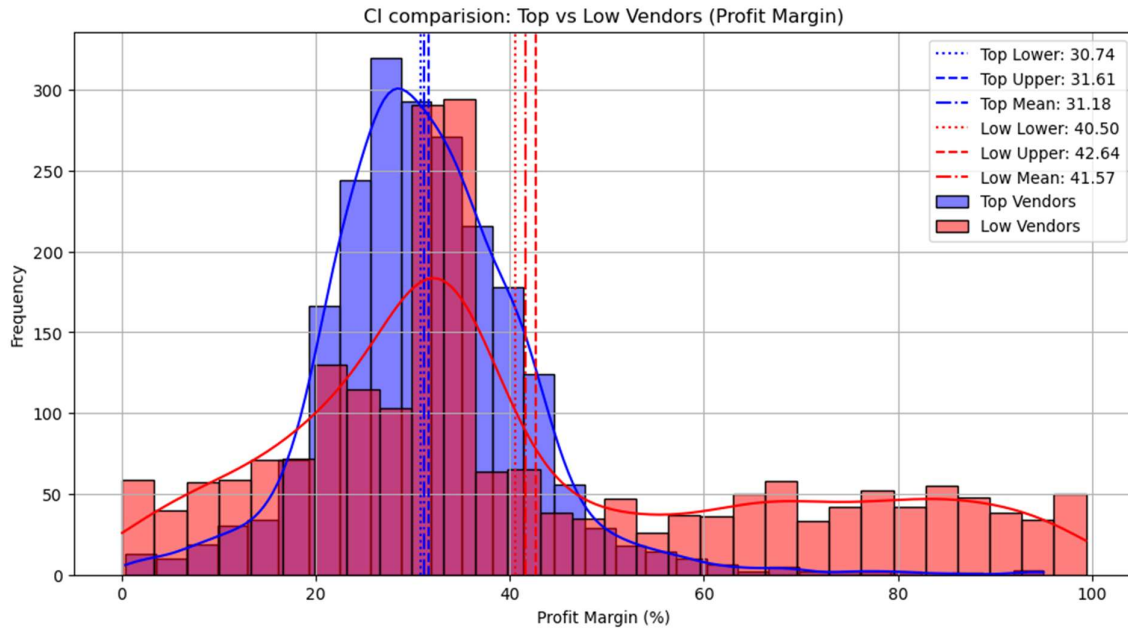
Top vendors’ Profit Margin (95% CI): (30.74), (31.61), Mean: 31.18

Low vendors’ Profit Margin (95% CI): (40.50), (42.64), Mean: 41.57

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.
- Low-performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.



6. Statistical Validation of Profit Margin Differences –

Hypothesis Testing:

H_0 (Null Hypothesis): There is no significant difference in the mean profit margins of top-performing and low-performing vendors.

H_1 (Alternative Hypothesis): There is a significant difference in the mean profit margins of top-performing and low-performing vendors.

Result: The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication: High-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on cost efficiency.

Final Recommendations:

- Re-evaluate pricing for low-sales, high-margin brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.

- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities or revising storage strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.